

Defensin gene family in *Medicago truncatula*: structure, expression and induction by signal molecules

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Abstract

A large gene family encoding the putative cysteine-rich defensins was discovered in *Medicago truncatula*. Sixteen members of the family were identified by screening a cloned seed defensin from *M. sativa* (Gao *et al.*, 2000) against the Institute for Genomic Research's (TIGR) *M. truncatula* gene index (MtGI version 7). Based on the comparison of their amino acid sequences, *M. truncatula* defensins fell arbitrarily into three classes displaying extensive sequence divergence outside of the eight canonical cysteine residues. The presence of Class II defensins is reported for the first time in a legume plant. *In silico* as well as Northern blot and RT-PCR analyses indicated these genes were expressed in a variety of tissues including leaves, flowers, developing pods, mature seed and roots. The expression of these genes was differentially induced in response to a variety of biotic and abiotic stimuli. For the first time, a defensin gene (TC77480) was shown to be induced in roots in response to infection by the mycorrhizal fungus, *Glomus versiforme*. Northern blot analysis indicated that the tissue-specific expression patterns of the cloned *Def1* and *Def2* genes differed substantially between *M. truncatula* and *M. sativa*. Furthermore, the induction profiles of the *Def1* and *Def2* genes in response to the signaling molecules methyl jasmonate, ethylene and salicylic acid differed markedly between these two legumes.

Abbreviations: Def, defensin; RT-PCR, reverse transcription-polymerase chain reaction; TC, tentative consensus sequence

Introduction

Antimicrobial proteins represent an important component of the innate defense response of plants. These proteins include thionins, lipid transfer proteins, hevein- and knottin-type proteins, snakins and defensins (Broekaert *et al.*, 1997; Garcia-Olmedo *et al.*, 1998). Plant defensins are among the earliest molecular effectors of innate immunity and are structurally and functionally related to insect and mammalian defensins

(Hoffmann *et al.*, 1999; Lehrer and Ganz, 2002; Thomma *et al.*, 2002). They are small, generally basic proteins of 45 to 54 amino acids and their tertiary structures share a common cysteine-stabilized α/β motif composed of three antiparallel β -sheets and one α -helix. Despite a similar backbone structure, a comparison of the primary amino acid sequences of over eighty defensins reveals that the conserved residues are restricted to the eight structurally important cysteines (Thomma *et al.*, 2002).

Plant defensins with diverse biological activity have been isolated from a number of dicotyledonous and monocotyledonous plants. A subset of these proteins inhibits the growth of a broad range of fungi at micromolar concentrations (Broekaert *et al.*, 1995; da Silva Conceicao and Broekaert, 1999; Almeida *et al.*, 2000; Gao *et al.*, 2000; Thomma *et al.*, 2002). Antifungal defensins are either morphogenic, causing reduced hyphal elongation with a concomitant increase in hyphal branching, or nonmorphogenic, causing reduced hyphal elongation without marked morphological distortion (Terras *et al.*, 1992; Broekaert *et al.*, 1995; Segura *et al.*, 1998).

Plant defensins have been isolated from a variety of tissues ranging from leaves, tubers, flowers, pods and seeds. Thus they are expressed during normal plant growth and development (da Silva Conceicao and Broekaert, 1999). Expression analysis of five *Arabidopsis* defensin genes indicates that while each organ expresses at least one defensin gene, some organs express two or more (Thomma and Broekaert, 1998). Among these, four are expressed constitutively and one is induced in leaves upon pathogen infection. A recent search of the fully sequenced *Arabidopsis* genome has revealed the presence of thirteen putative defensin genes and two genes containing a plant defensin domain (Thomma *et al.*, 2002). Therefore, much remains to be learned regarding the expression of several members of the defensin gene family in *Arabidopsis*.

In situ hybridization and immunocytochemical studies have shown that defensins are expressed primarily in the peripheral cell layers of different organs. In radish, the highest concentrations of defensins occur in the outer cell wall of the epidermal cells of cotyledons, hypocotyls, and endosperm (Terras *et al.*, 1995). In sugar beets, they are located primarily in the xylem, stomatal cells and in cell walls lining the sub-stomatal cavities (Kragh *et al.*, 1995). In *Nicotiana glauca*, expression of the flower-specific defensin gene NaD1 occurs in the outermost cell layers of petals, sepals and styles (Lay *et al.*, 2003a, b). These observations are consistent with a role of defensins in protection against pathogens. Furthermore, in plants such as radish and *Arabidopsis*, specific defensin genes are induced in vegetative tissues upon challenge by pathogens. The induction of these genes is systemic with expression in both

infected and uninfected leaves (Terras *et al.*, 1995; Penninckx *et al.*, 1996). The data available so far indicate that the members of the defensin gene family are expressed differentially not only during development, but also in response to pathogen infection. Apart from the isolation of defensin from the seed of *Medicago sativa* (Gao *et al.*, 2000), little is known regarding the tissue-specific and stress-inducible expression of defensin genes in *M. sativa* and *M. truncatula*.

It is well established that signaling pathways controlling plant responses to pathogen infection operate through the action of phytohormones such as salicylic acid (SA), jasmonate (JA) and ethylene. Interplay between signals allows the plant to fine-tune local and systemic defense responses (Feys and Parker, 2000). *Arabidopsis* mutants defective in ethylene or JA pathways have demonstrated that activation of the defensin gene, *PDF1.2*, by the fungal pathogen *Alternaria brassicicola* requires concomitant activation of both the ethylene and JA signaling pathways and that this activation is independent of SA (Penninckx *et al.*, 1998). In this study, expression of the *PDF1.2* gene was completely abolished in the ethylene and JA response mutants, *ein2* and *coi1*, respectively. Recently, a transcription factor, Ethylene Response Factor1 (ERF1), has been shown to be responsible for the ethylene/JA-dependent activation of the *PDF1.2* gene in *Arabidopsis* (Lorenzo *et al.*, 2003). Constitutive expression of ERF1 was sufficient to activate *PDF1.2* in the *coi1* and *ein2* mutant backgrounds (Solano *et al.*, 1998; Lorenzo *et al.*, 2003). To what extent ethylene, JA and SA signaling pathways contribute to the activation of defensin genes in either *M. truncatula* or in *M. sativa* remains to be determined.

In this report, we have undertaken a study aimed at determining the structure and expression of defensin genes in a model legume *M. truncatula*, which has emerged as a model plant for genomic approaches to study plant-microbe symbiotic associations (Cook, 1999; Harrison, 2000; Oldroyd and Geurts, 2001; Thoquet *et al.*, 2002). *M. truncatula* possesses a small genome, is diploid, has a fast generation time and is transformable (Blondon *et al.*, 1994; Chabaud *et al.*, 2003). Currently, both structural and functional genomic approaches are being pursued for *M. truncatula* (www.medicago.org). We have previously reported the isolation and cloning of a cDNA encoding an

antifungal defensin present in the seed of *M. sativa* (Gao *et al.*, 2000). Ectopic overexpression of this defensin in transgenic potato conferred strong resistance to the fungal pathogen *Verticillium dahliae* and resistance was sustainable under field conditions (Gao *et al.*, 2000). In the present study, we have screened The Institute for Genomic Research (TIGR) *M. truncatula* gene index (MtGI release 7.0) to identify a defensin gene family consisting of at least sixteen members. These defensins fall into three classes that display extensive sequence divergence outside of the eight conserved cysteines. *In silico*, RT-PCR and northern blot analyses indicate that these genes are expressed in a variety of tissues and are also induced in response to a variety of biotic and abiotic stimuli. We further show that the tissue-specific expression of the cloned defensin genes differs markedly in *M. truncatula* and *M. sativa*. In addition, expression of these genes in response to ethylene, JA and SA also differs between these two legumes.

Materials and methods

Cloning and sequence analysis of Def1 and Def2 genes from M. truncatula and M. sativa

The cloning and nucleotide sequence analysis of an *M. sativa* full-length cDNA clone encoding a seed defensin, MsDef1.6, previously referred to as alfAFP, have been published previously (Gao *et al.*, 2000). For cloning the homologues of the *MsDef1.6* gene from *M. truncatula*, genomic DNA was used as a template for PCR amplification. The forward primer, 5' ATGGAGAAGAAATCAGTAGC 3' and a reverse primer, 5' TTAACA TCTTTTAGTACACC 3' corresponding to the 5' and 3' ends of the *MsDef1.6* coding sequence were used for PCR amplification. Typical PCR conditions were 30 cycles with a 90 s, 94 °C denaturation step; a 30 s, 50 °C annealing step; and a 60 s, 72 °C extension step. One PCR product of 318 bp was obtained. The same primers were used for amplification of the *MsDef1* genes from the genome of *M. sativa* variety LegenDairy. Six PCR products ranging in size from 219 to 793 bp from *M. sativa* were obtained. All PCR products were cloned into the pCR2.1-TOPO vector (Invitrogen, Carlsbad, CA) and two inde-

pendent clones for each PCR product were sequenced on an ABI Prism 377 at Iowa State University.

A search of TIGR's *M. truncatula* Gene Index MtGI (release 4.0) (<http://www.tigr.org/tdb/tgi/mtgi>) using the *MsDef1.6* cDNA sequence as a query yielded one tentative consensus sequence (TC50237) of 483 bp with 75% nucleotide identity which we named *MtDef2.1*. A cDNA clone from this TC was generously provided by The Samuel Roberts Noble Foundation, Ardmore, OK and sequenced. For the PCR amplification of genomic fragments from *M. sativa* and *M. truncatula*, the forward primer, 5' ATGGAGAAGAAATCAATAGCT 3' and the reverse primer, 5' TTAACAG TTTTGAGTGCAGA 3', corresponding to the 5' and 3' ends of the *MtDef2.1* sequence were used. Three PCR products of ~2200 bp, 1300 bp, and 563 bp from *M. sativa* DNA and one PCR product of 897 bp from *M. truncatula* DNA were cloned into the pCR2.1-TOPO vector and sequenced.

In silico identification of defensin family members

The coding sequences of *MsDef1.6*, *MtDef1.1* and *MtDef2.1* were used as BLASTN and TBLASTX (Altschul *et al.*, 1997) queries for TIGR's MtGI (release 7.0). Sequence hits with *E*-values less than 10^{-4} were identified as putative defensin homologues. In order to identify additional homologues, TBLASTX was used to query the newly identified defensin homologues against MtGI, using an *E*-value cutoff of 10^{-4} . Reiterative TBLASTX analyses were continued until no further homologues could be identified. Sequence alignments of the 20 TCs and singletons were used to verify defensin similarity. Further nucleotide alignments in Vector NTI Suite 8.0 (InforMax, Inc., Frederick, Maryland) reclassified some singletons as members of established TCs or as corresponding to our cloned defensin genes. To identify genomic sequences corresponding to this family of defensins, the nucleotide sequence of each member was used as a BLASTN query against the available *M. truncatula* genome sequence. Sequences with 99% or more nucleotide identity were considered matches to the *M. truncatula* genome. Alignments of the deduced amino acid sequences of the *M. truncatula* defensins were done using ClustalW (Thompson *et al.*, 1994) with the blosum62mt2 scoring matrix (Henikoff and Henikoff, 1993) in

Vector NTI. Sequencing of selected ESTs and genomic clones was conducted at Iowa State University on an ABI Prism Model 377. Signal peptide cleavage sites were predicted with SignalP (Nielsen *et al.*, 1997) and intron splice sites were predicted by alignment with the EST and *MsDefl.6* cDNA sequences (Gao *et al.*, 2000).

Plant growth conditions and treatment with ethylene, MeJA and SA

M. truncatula 'Jemalong' A17 seeds were scarified for 10 min in sulfuric acid, surface-sterilized in 20% bleach plus Tween (Sigma, St. Louis, MO), germinated on water agar for 14 days and were then planted in a vermiculite mix. Seeds of *M. sativa* variety Legend were planted directly in the vermiculite mix. Seeds of *Arabidopsis* ecotype Columbia were sown into Scotts MetroMix200 and kept in a propagator for 4 days at 4 °C to ensure even germination. The pots of germinated seeds were then moved to a greenhouse with a temperature of 22 °C and a 12-h light/12-h dark cycle.

Plants were treated with ethylene or SA 14–21 days after planting as described (Samac and Shah, 1991). For ethylene treatment, plants were sprayed to runoff with ethephon (2-chloro-ethylphosphonic acid) and enclosed in plastic bags. For SA treatment, plants were sprayed to run-off with 1 mM, 2.5 mM, 5 mM or 10 mM solution of SA, pH 6.5. Plants were exposed to MeJA by applying three 5 µl drops of a 100 µM solution to each leaf (Sigma, St Louis, MO). In all cases, the *Arabidopsis* plants, used as controls for hormone treatments, were treated in the same manner as for chemically treated plants but without the addition of MeJA, ethylene or SA. Plants were harvested and frozen in liquid nitrogen at 48, 72 or 96 h after treatment for isolation of RNA.

Infection of M. truncatula with mycorrhizal fungus Glomus versiforme

Medicago truncatula 'Jemalong' A17 seeds were surface-sterilized and grown in sterile Turface in 11 cm diameter plant pots in growth rooms under a 16-h light, 8-h dark cycle at 25 °C/22 °C. Following the development of the first trifoliate leaf, the plants were inoculated with

G. versiforme spores (5000 spores per pot) or mock-inoculated with the filtrate from the spore sterilization procedure as described previously (Harrison and Dixon, 1993; Harrison *et al.*, 2002; Liu *et al.*, 2003). Plants were harvested at 1, 2, 3 and 4 weeks post-inoculation and the percent root length colonized was assessed by the grid-line intersect method (McGonigle *et al.*, 1990). Root samples showed 11%, 45%, 53% and 73% root length colonized at weeks 1, 2, 3 and 4, respectively.

RT-PCR analysis of RNA from

G. versiforme-infected M. truncatula roots

RNA was prepared from *G. versiforme*-infected roots according to (Chomczynski and Sacchi, 1987). RT-PCR analyses were carried out as described previously (Liu *et al.*, 2003). Briefly, reverse transcription reactions contained 5 µg of RNA in 40 µl reactions with Superscript II reverse transcriptase (Invitrogen Life Technologies) according to the manufacturer's protocol. PCR reactions contained 1–2 µl of reverse transcription reaction (adjusted based on the *EF1* levels in a 20 µl PCR with Promega *Taq* polymerase). The following primers were used: TC77480-Forward 5'ACACTATCTTTCTGTTT CTCAGCC 3' and Reverse 5' CAATTGA AGTAGCAGAAGCAAG; TC85327-Forward 5'ATGGTGGCAGAAGCAAGAA 3' and Reverse 5' CAAAAGCATCTACGACGG 3'. These primers were designed to generate RT-PCR products of 211 and 143 bp, respectively. The number of cycles was adjusted to remain in the linear amplification range and was as follows: TC77480 24 cycles, TC85327 25 cycles, *MtPT4* 24 cycles, *EF1a* 23 cycles. Annealing temperatures were 50 °C for TC77480, 48 °C for TC 85327, 55 °C for *MtPT4*, and 55 °C for *EF1a*.

RNA extraction and RNA gel blot analysis

Newly emerged young leaves, fully expanded mature leaves, roots and flowers from 75-day-old *M. truncatula* and 80-day-old *M. sativa* plants were excised, immersed in liquid nitrogen and stored at –80 °C. Total RNA was extracted from

the frozen tissues using the Trizol reagent (Invitrogen, Carlsbad, CA) according to the manufacturer's recommendations. Total RNA from dry seed was isolated as described (Vicent and Delseny, 1999). RNA pellets from all samples were dissolved in RNase-free water and quantified spectrophotometrically. Ten micrograms of RNA from each sample was fractionated by electrophoresis on 1.4% agarose gels containing formaldehyde, and blotted onto Hybond-XL nylon membrane (Amersham Pharmacia Biotech). The membranes were baked at 75 °C for 2 h and incubated in hybridization buffer [50% formamide, 5 × Denhardt's solution (1× Denhardt's solution is 0.02% ficoll, 0.02% polyvinylpyrrolidone, and 0.02% BSA), 1% SDS and 200 µg/ml tRNA] at 42 °C for 4–6 h. Radioactive ³²P-dCTP-labeled probes were generated by random priming according to manufacturer's recommendations (Prime-It RmT Random Primer Labeling Kit, Stratagene, CA) using the PCR-generated DNA fragments encoding the mature MtDef1 and MtDef2 proteins. The gene-specific probes corresponding to the 3' flanking sequences of the *MsDef1.6* and *MtDef2.1* genes were obtained by PCR. The probes were heat denatured and incubated with the blocked membrane overnight at 42 °C. Membranes were washed with 1 × SSC, 0.2% SDS for 20 min once at room temperature and three times at 50 °C and then exposed to the X-ray films.

Results

Cloning and sequence analysis of genes encoding Def1 and Def2 defensins

The full-length cDNA clone encoding seed defensin, MsDef1.6, from *M. sativa* has been cloned and sequenced previously (Gao *et al.*, 2000). In order to perform a comparative analysis of the *Def1* genes from *M. truncatula* and *M. sativa*, PCR amplification was used to identify the *Def1* homologues from both *M. truncatula* and *M. sativa*. Six PCR amplification products ranging in size from 219 to 793 bp were obtained from the *M. sativa* genomic DNA and sequenced. *MsDef1.1*, *MsDef1.3*, *MsDef1.4*, *MsDef1.5* and *MsDef1.6* genes contain an intron, whereas *MsDef1.2* gene does not (Figure 1A). Each of these genes encodes a putative

27-amino acid signal peptide and a 45-amino acid mature protein. All six proteins are nearly identical in their amino acid sequences. As shown in Figure 1B, MsDef1.3, MsDef1.4, MsDef1.5 and MsDef1.6 mature proteins have arginine at position 1, whereas MsDef1.1 and MsDef1.2 have lysine. Surprisingly, nucleotide sequence of the 318 bp *MsDef1.1* was identical to that of *MtDef1.1* from *M. truncatula*. Both genes contain a 99 bp intron with an identical sequence. Thus, the *MtDef1.1* gene is an identical homologue of the *MsDef1.1* gene of *M. sativa*. Although *M. sativa* and *M. truncatula* genomes are closely related (Zhu *et al.*, 2002), the perfect identity between these two genes is surprising. From these data, it is clear that *M. sativa* contains at least six *Def1* genes, whereas *M. truncatula* contains only one gene, *MtDef1.1*. A Southern blot analysis of *M. truncatula* genomic DNA was performed using the DNA fragment corresponding to the mature MtDef1.1 protein as a probe. The DNA sequence encoding signal peptide was excluded from the hybridization probe because it hybridized with other genes unrelated to defensins (data not shown). *MtDef1.1* probe hybridized strongly with a 9.5 kb *HindIII* fragment and weakly with a 4.9 kb *HindIII* fragment. A similar pattern was observed in lanes containing the *BglIII* and *EcoRI* digests of genomic DNA. Since these restriction sites are not present within the *MtDef1.1* coding sequence, we conclude that there is one more *MtDef1.1*-related gene in the *M. truncatula* genome (Figure 2). The differences in primer annealing sites may have prevented the amplification of the second *MtDef1.1*-like gene by PCR amplification of the genomic DNA.

A search of the TIGR *M. truncatula* Gene Index (MtGI release 4.0) using the *MsDef1.6* cDNA sequence as a query yielded one singleton EST clone of 483 bp (GenBank accession number BF633403) with 75% identity at the nucleotide sequence level to the cDNA clone. This EST clone was derived from the cDNA library, NF-DT, constructed from mRNA isolated from drought-stressed plants and was obtained from The Samuel Roberts Noble Foundation (Ardmore, OK). The nucleotide sequence of this clone, designated *MtDef2.1*, is predicted to encode a protein with a 27-amino acid signal peptide and a mature protein of 46 amino acids. The signal peptide sequences of MtDef1.1 and MtDef2.1 are remarkably conserved, with only 3 amino acid differences

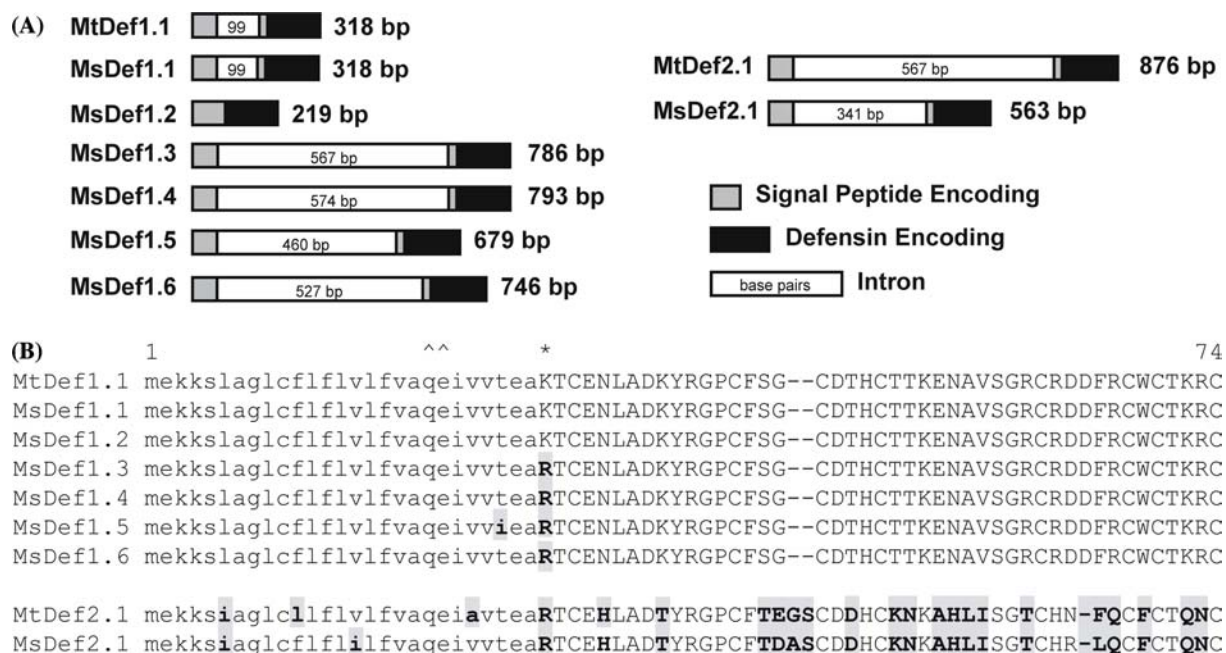


Figure 1. Comparison of the gene structure and deduced amino acid sequences of *Def1* and *Def2* defensin genes from *M. truncatula* and *M. sativa*. A. Exon-intron maps of the *Def1* and *Def2* genes are shown. B. The deduced amino acid sequences of the putative signal peptides and mature proteins. First amino acids of the predicted mature proteins are shown by an asterisk. First amino acid of MsDef1.6 was determined by amino acid sequence analysis of the protein purified from the seed of *M. sativa* (Gao *et al.*, 2000). Carrots denote the position of an intron. The accession numbers for gene sequences are: MsDef1.6, AY681967; MsDef1.3, AY681970; MsDef1.4, AY681969; MsDef1.5, AY681968; MsDef1.6, AY681967; MsDef1.1, AY681972; MtDef1.1, AY313166; MsDef2.1, AY681973; MtDef2.1, AY313169.

(Figure 1B). However, the sequence alignment shows that the mature proteins have diverged significantly, differing in 18 out of 44 amino acids. MtDef2.1 differs from the recently published sequence of the *Pisum sativum* defensin protein, Psd1, in only 6 amino acids (Almeida *et al.*, 2002).

DNA fragments containing the *Def2* gene were PCR amplified from the genomic DNA of *M. truncatula* as well as the *M. sativa* variety LegenDairy using primers specific for *MtDef2.1*. One PCR product of 876 bp was obtained from *M. truncatula* genomic DNA (Figure 1A). The nucleotide sequence of the 876 bp product from *M. truncatula* corresponded to that of the *MtDef2.1* cDNA clone in the protein coding region of the gene, designated *MtDef2.1*, and contained a 654 bp intron which occurred in the same position as in the *MtDef1.1* gene. Of the three PCR products obtained from the *M. sativa* genomic DNA, only the 563 bp product encoded defensin. The nucleotide sequences of the 1300 bp

and the 2200 bp products revealed no significant homology with the *MtDef2.1* cDNA sequence and lacked the eight conserved cysteines. The 563 bp gene, designated *MsDef2.1*, was interrupted by a 341 bp intron. The predicted MsDef2.1 and MtDef2.1 defensin sequences differ from each other in 4 amino acids. Thus, we were able to amplify only one *Def2* gene from *M. sativa* and *M. truncatula*. The *MtDef2* probe hybridized with only one fragment in each of the three restriction digests of the *M. truncatula* genomic DNA (Figure 2).

The presence of a large gene family encoding defensin in M. truncatula

Using the nucleotide sequences of *MtDef1.1* and *MtDef2.1* as queries, a screen of TIGR MtGI database (MtGI release 7.0) containing 189,919 ESTs originating from 43 cDNA libraries was performed with successive rounds of BLASTN

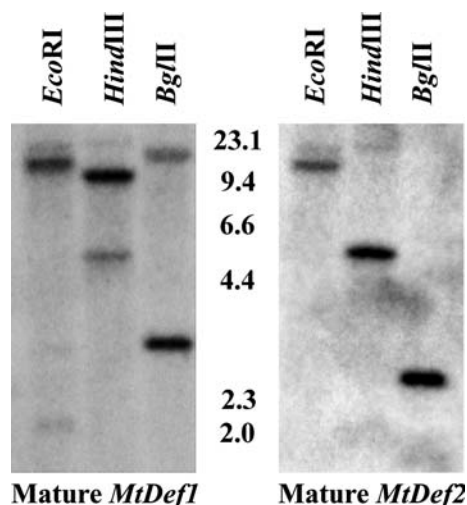


Figure 2. Southern blot analysis of *M. truncatula* DNA for *MtDef1* and *MtDef2* genes. Five microgram of genomic DNA digested with *EcoRI* (lane 1), *HindIII* (lane 2), and *BglII* (lane 3), was transferred to Hybond-XL membrane and hybridized with 32 P-labeled *MtDef1* and *MtDef2* probes as described in 'Materials and methods'. Size markers are indicated.

and TBLASTX. This screen led to a discovery of 18 putative defensin genes represented by 10 tentative consensus (TC) sequences and eight singletons (unique non-overlapping sequences). Careful nucleotide alignments revealed that some of the original singletons might indeed be members

of already established TCs, and the family has been currently sized to 16 consisting of 10 TCs and 6 singletons. Figure 3 shows the summary of ESTs for the *M. truncatula* TCs and singletons with 100% identical residues between all family members shown in black and those present in greater than 40% of the ESTs in grey. Based on their amino acid sequence alignment, the putative defensin sequences were arbitrarily divided into three classes. The amino acid sequences of six defensins (BG452001, TC85327, CA990416, TC89352, AJ498901, TC82368) in class I share 46 to 96% similarity (Table 1). Three defensins (TC76542, TC87634, TC90828) in class II share 35 to 52% similarity. However, between the fifth and the sixth conserved cysteine residues, all three defensins have only 4 amino acids. Further, TC76542 and TC87634 encode proteins that contain a deletion of two amino acids between the last two conserved cysteines, whereas TC90828 encodes a protein with a carboxy-terminal extension of three amino acids (Asn–Gly–Asp) not present in other defensins. The only plant defensin (Bp4A) that carries a deletion of two amino acids between the last two conserved cysteines has been reported to be a pollen-coating protein in *Brassica napus* (Albani *et al.*, 1990), whereas defensins with a carboxy-terminal extension have been reported in several plants (da Silva Conceicao and Broekaert,



Figure 3. The defensin multigene family of *M. truncatula*. Alignment of 16 defensins sequences is shown. Identical residues conserved in all defensins are in black background, whereas those residues that are conserved in at least 40% or more defensins are in lower case. The predicted signal peptide sequences are shown in *lower case*. Consensus sequence is based on the identical residues conserved in all Medicago defensins.

[illegible]

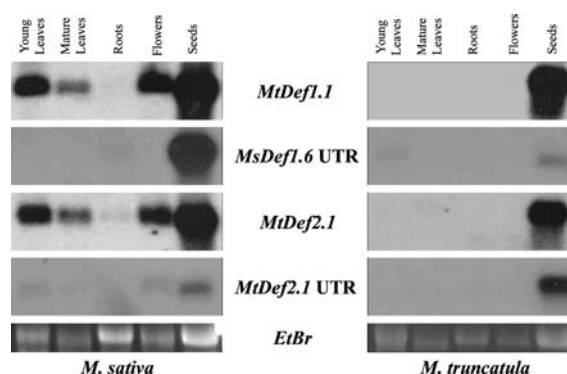


Figure 4. Northern blot analysis of *Def1* and *Def2* gene expression in *M. truncatula* and *M. sativa*. Ten microgram of total RNA isolated from the indicated tissues was run in each lane, blotted and hybridized with the probes indicated. Equal loading of RNA was assessed by ethidium bromide staining of the 18S ribosomal RNA.

from the EST clone is able to detect *MsDef2.1* transcripts in seed indicating some homology between the 3' flanking sequences of the *MsDef2.1* and *MtDef2.1* genes. Collectively, the results show that the expression patterns of these genes in different organs of *M. sativa* differ substantially from those in *M. truncatula*.

In silico expression analysis of defensin genes from M. truncatula

Since each putative defensin TC sequence is derived from individual overlapping ESTs, it represents a unique defensin transcript expressed in different libraries but presumably from a single gene. Differences in the number of ESTs for each defensin TC likely reflect differences in the abundance of transcripts from the corresponding gene. It is clear that the defensin genes represented by these TCs and singletons are expressed in different tissues at different stages of development and also in response to a variety of biotic and abiotic stimuli (Table 2). Multiple defensin genes are expressed in certain tissues. For example, transcripts of the genes represented by TC82368, TC85327 and TC89352 are present in the library from seeds 25 to 35 days post-pollination. Some genes (TC76542, TC82368, TC85327) are expressed in more than one tissue.

It is noteworthy that none of these defensin genes are represented in the cDNA libraries of normal root tissue. However, some of these genes are induced moderately or highly in roots in response to phosphate starvation, mycorrhizal fungal infection or rhizobacterial infection. For example,

TC85327 and TC77480 are each represented by 7 ESTs in the cDNA library of roots 15-day post-inoculation with the mycorrhizal fungus, *Glomus intraradicis*. In order to test if these genes are induced in roots during colonization by another mycorrhizal fungus, *G. versiforme*, we performed RT-PCR analysis of RNA isolated from roots inoculated with fungal spores at 1, 2, 3 and 4 weeks post-inoculation. TC77480 was strongly induced in roots 2, 3 and 4 weeks post-inoculation, whereas TC85327 was constitutively expressed in mock-inoculated as well as *G. versiforme*-infected roots (Figure 5). The induction of TC77480 was nearly as strong as that observed for the phosphate transporter gene *MtPT4* (Harrison *et al.*, 2002).

TC87634 is represented by 9 ESTs in a cDNA library of nodules 4-day-post-inoculation with *Sinorhizobium meliloti*. These are the first examples of defensin genes that are induced during plant–fungal or plant–bacterial symbiosis. Only one gene (TC85327) seems to be induced weakly in roots in response to a fungal pathogen, *Phytophthora medicaginis*. Surprisingly, no ESTs were identified from the cDNA libraries derived from *Colletotrichum trifolii*- or *Phoma medicaginis*-infected plants, although some (TC85327 and TC87273) do respond to elicitors prepared from the cell wall of *P. sojae* or yeast. TC85327 and TC76542 also respond to multiple abiotic stimuli such as drought stress and ultraviolet light irradiation. AJ498901 representing the *MtDef1.1* gene was identified as a single EST clone from a cDNA library of developmentally pooled pods containing seeds (Table 2). On the other hand, TC82368, which represents the *MtDef2.1* gene, consists of three

Table 2. Expression summary for putative *M. truncatula* defensin-encoding ESTs.

TC or EST	Library ^a	No of ESTs ^b	% of library	<i>Medicago truncatula</i> : tissue, developmental stage, growth conditions ^a
TC76542	NF-IR	6	0.12	UV and gamma irradiated 21-day-old seedlings
	NF-LF	6	0.07	Very young, developing, mature and senescing leaves
	NF-DT	4	0.05	Drought-stressed whole plantlets
	MTAMP	1	0.03	Mycorrhizal roots – 35 dpi with <i>Glomus intraradices</i>
	MTPOSE	1	0.09	Developmentally pooled pods with seeds
	NF-PL	1	0.01	Phosphate-starved leaves from mature plants
TC77480	MTGIM	17	1.01	Mycorrhizal roots – 15 dpi with <i>Glomus intraradices</i> – subtracted library
	MtBC	7	0.08	Mycorrhizal roots – 15 dpi with <i>Glomus intraradices</i>
	MTAMP	1	0.03	Mycorrhizal roots – 35 dpi with <i>Glomus intraradices</i>
TC82368	GLSD	1	0.02	Late seeds – 25 to 35 dap
	NF-DT	1	0.01	Drought-stressed whole plantlets
	NF-LF	1	0.01	Very young, developing, mature and senescing leaves
TC84759	KV1	1	0.04	Nodulated roots – 1 dpi with <i>Sinorhizobium meliloti</i>
	MTAMP	1	0.03	Mycorrhizal roots – 35 dpi with <i>Glomus intraradices</i>
TC85327	NF-IR	10	0.19	UV and gamma irradiated 21-day-old seedlings
	NF-DT	9	0.1	Drought-stressed whole plantlets
	MtBC	7	0.08	Mycorrhizal roots – 15 dpi with <i>Glomus intraradices</i>
	NF-PL	7	0.07	Phosphate-starved leaves from mature plants
	MGHG	4	0.15	Seedling roots after treatment with beta-glucan elicitor preparation from <i>Phytophthora sojae</i>
	GESD	4	0.09	Early seeds – 11 to 19 dap
	MHRP-	4	0.09	Phosphate-starved roots – 30 days
	NF-LF	4	0.05	Very young, developing, mature and senescing leaves
	GVN	3	0.05	Nitrogen-fixing root nodules – 30 dpi with <i>Sinorhizobium meliloti</i>
	GVSN	3	0.11	Senescent nodules – 40 to 60 dpi with <i>Sinorhizobium meliloti</i>
TC87273	DSIR	2	0.08	Roots 10 dpi with <i>Phytophthora medicaginis</i>
	KV1	1	0.04	Nodulated roots – 1 dpi with <i>Sinorhizobium meliloti</i>
	MHAM	1	0.01	Colonized roots – 10 to 38 dpi with <i>Glomus versiforme</i>
	MTFLOW	1	0.08	Developmentally pooled flowers
	MTPOSE	1	0.09	Developmentally pooled pods with seeds
	BNIR	1	0.03	3-week-old roots 3 dpi with the nematode <i>Meloidogyne incognita</i>
	GLSD	1	0.02	Late seeds – 25 to 35 dap
	NF-FL	1	0.02	Very young, developing, fully opened flowers and flowers in early transition into pods
	NF-EC	8	0.07	Root-derived cell cultures elicited with yeast cell wall extract 0.5 to 24 h after induction
	MtBA	1	0.01	Nitrogen-starved root tips
	MtBB	1	0.01	Root nodule 4 dpi with <i>Sinorhizobium meliloti</i>
	MHAM	1	0.01	Colonized roots – 10 to 38 dpi with <i>Glomus versiforme</i>
	MGHG	1	0.04	Seedling roots after treatment with beta-glucan elicitor preparation from <i>Phytophthora sojae</i>
	MtBB	9	0.12	Root nodule 4 dpi with <i>Sinorhizobium meliloti</i>
	6KUG	1	0.02	Mixed tissues various stages
TC89352	MTPOSE	2	0.17	Developmentally pooled pods with seeds
	GLSD	2	0.04	Late seeds – 25 to 35 dap
TC90828	MtBC	2	0.02	Mycorrhizal roots – 15 dpi with <i>Glomus intraradices</i>
	MTGIM	1	0.06	Mycorrhizal roots – 15 dpi with <i>Glomus intraradices</i> – subtracted library
TC90864	MtBC	2	0.02	Mycorrhizal roots – 15 dpi with <i>Glomus intraradices</i>
	MTGIM	1	0.06	Mycorrhizal roots – 15 dpi with <i>Glomus intraradices</i> – subtracted library
AJ498901	MTPOSE	1	0.09	Developmentally pooled pods with seeds
AJ499268	MTGIM	1	0.06	Mycorrhizal roots – 15 dpi with <i>Glomus intraradices</i> – subtracted library
AJ499425	MTGIM	1	0.06	Mycorrhizal roots – 15 pi with <i>Glomus intraradices</i> – subtracted library
AL385826	MtBC	1	0.01	Mycorrhizal roots – 15 dpi with <i>Glomus intraradices</i>
BF649584	NF-EC	1		Root-derived cell cultures elicited with yeast cell wall extract 0.5 to 24 h after induction
BF650436	NF-EC	1		Root-derived cell cultures elicited with yeast cell wall extract 0.5 to 24 h after induction
BG452001	NF-DT	1	0.01	Drought-stressed whole plantlets
BQ1 24948	GLSD	1	0.02	Late seeds – 25 to 35 dap
CA990416	GESD	1	0.02	Early seeds – 11 to 19 dap

^a All library information available at the *Medicago truncatula* Consortium website, at www.medicago.org.^b Number of ESTs from each library that correspond to this nucleotide sequence.^c DPI, days post-inoculation.^d DAP, Days after pollination.

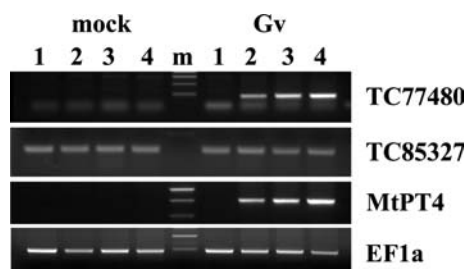


Figure 5. RT-PCR analysis of TC77480 and TC85327 gene expression in mock-treated or *G. versiforme*-infected roots of *M. truncatula* 1-, 2-, 3-, 4-week after inoculation. MtPT4 gene was used as a positive control for *G. versiforme*-induced expression. EF1a was used as a constitutively expressed gene; 'm' denotes size markers.

ESTs identified from three independent cDNA libraries of seed (25 to 35 days after pollination), drought-stressed whole plants and leaves (Table 2). Using northern blot analysis, high expression of the *MtDef2.1* gene was detected in dry mature seed, but not in leaves (Figure 4). Since the RNA from the senescing leaf tissue was not used for this analysis, the possibility remains that the EST clone is derived from the cDNA library of the senescing leaf tissue. Alternatively, a more sensitive RT-PCR approach may be needed to detect low level expression in the leaf tissue.

Differential induction of Def1 and Def2 gene expression in response to signaling molecules

Several plant defense genes are activated in response to endogenous signaling molecules including SA, MeJA and ethylene. It has been previously reported that the Arabidopsis defensin gene, *PDF1.2*, is induced by exogenous treatment with either ethylene or MeJA, but not by SA (Penninckx *et al.*, 1998). We tested the induction of the *Def1* and *Def2* genes in response to these signal molecules in *M. sativa* and *M. truncatula* seedlings and compared it with the induction of the Arabidopsis *PDF1.2* gene under the same conditions. RNA was extracted from two to 3-week-old seedlings exposed to ethylene, MeJA or SA or from control seedlings treated similarly without the signaling molecule. RNA blot analysis indicated that neither *MtDef1.1* nor *MtDef2.1* was induced in *M. truncatula* seedlings by ethylene treatment. However, in *M. sativa*, basal expression of the *MsDef1* gene(s) and the *MsDef2.1* gene was

strongly down-regulated by ethylene (Figure 6A). A strong induction of the Arabidopsis defensin *PDF1.2* gene by ethylene was observed in this experiment as reported previously (Penninckx *et al.*, 1998). In contrast to ethylene, treatment of *M. truncatula* seedlings with MeJA led to a strong induction of the *MtDef1.1* and *MtDef2.1* genes (Figure 6B). Under our experimental conditions, weak induction of these genes after 48 h of treatment and strong induction after 72 h of treatment was observed. This induction was significantly reduced after 96 h of treatment. In *M. sativa*, there is a high basal level of the expression and no further induction by MeJA of either the *MsDef1* gene(s) or the *MsDef2.1* gene. As expected, Arabidopsis *PDF1.2* defensin gene was also strongly induced by MeJA after 48 h of treatment and weak induction after 72 h of treatment. Under the conditions of our greenhouse, we have consistently observed basal expression of the *PDF1.2* gene in Arabidopsis. We have observed weak induction of the *MtDef1.1* and *MtDef2.1* genes only after treatment of the *M. truncatula* seedlings with either 5 mM or 10 mM SA, but not with 1 mM or 2.5 mM SA (Figure 6C). No induction of these genes by SA was observed in *M. sativa*. These results demonstrate that MeJA, but not ethylene or SA, is likely a signaling molecule playing a role in the activation of defensin genes in *M. truncatula*; however, in *M. sativa*, *MsDef1* gene(s) and *MsDef2.1* genes are constitutively expressed, but they are responsive to ethylene signaling.

Discussion

A defensin protein, MsDef1.6, was previously isolated from the *M. sativa* seed on the basis of its strong *in vitro* antifungal activity against certain plant fungal pathogens and the gene encoding this protein was demonstrated to confer strong resistance to the fungal pathogen *V. dahliae* when overexpressed in transgenic potato (Gao *et al.*, 2000). This provided strong evidence that a plant defensin was able to restrict fungal growth *in planta*. In order to build on this important knowledge base, we have systematically begun to identify and study defensin gene expression and function in a model legume, *M. truncatula*, a close relative of the forage crop *M. sativa*.

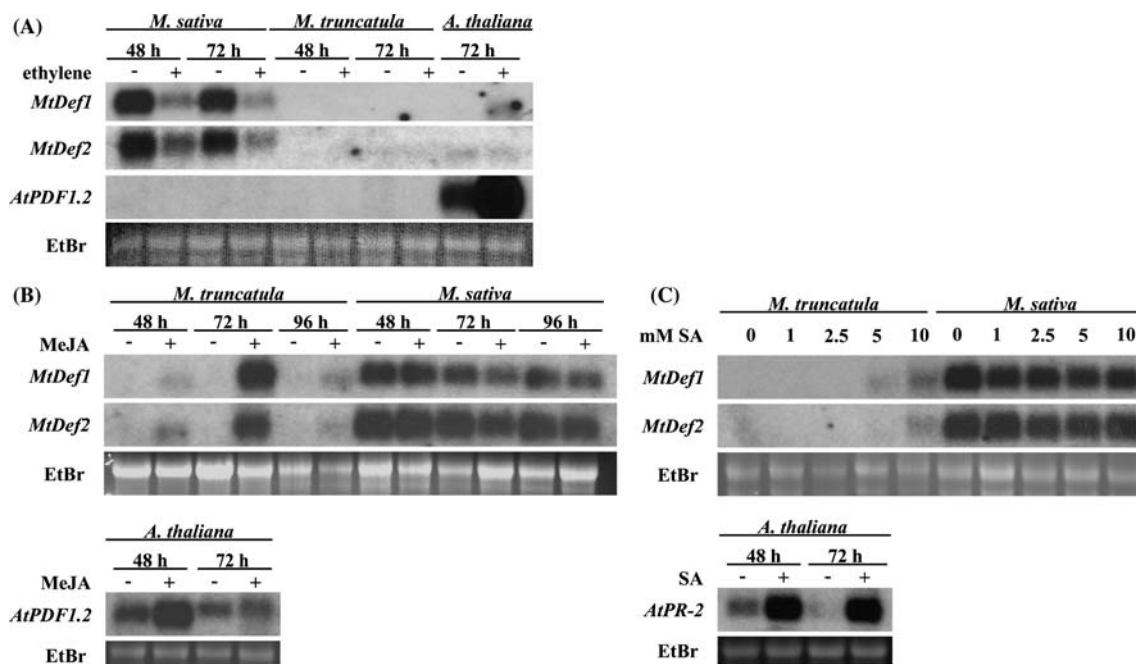


Figure 6. Induction of defensin gene expression by the signaling molecules ethylene, MeJA and SA in *M. truncatula*, *M. sativa* and *A. thaliana*. Total RNA was isolated from 2-week-old seedlings that were either treated with (A) ethylene, (B) MeJA and (C) SA or mock-treated as described in Materials and Methods. Blots were hybridized with the *MtDef1.1*, *MtDef2.1* and *AtPDF1.2* probes as indicated. *AtPR-2* was used as a positive control for SA-induced gene expression in *A. thaliana*.

In this report, we have employed *in silico* as well as molecular biological approaches to identify and categorize defensin gene sequences in *M. truncatula*. TIGR MtGI (release 7.0) contains 189,919 ESTs derived from a collection of libraries constructed from normal tissues and tissues exposed to a number of biotic and abiotic stimuli. The ESTs have been clustered into tentative consensus sequences producing a set of unique virtual transcripts made of both TCs and singletons. We have discovered a large gene family of 16 members encoding putative defensins in *M. truncatula*. It is likely that additional defensin genes will be discovered and that the exact size of the defensin gene family will be known only when the complete sequence of the genome becomes available. Recently, a gene family encoding putative defensins has also been discovered in Arabidopsis (Thomma *et al.*, 2002). An analysis of the genome sequence has revealed the presence of 13 putative defensin genes and 2 additional genes encoding defensin fusions. Based on the phylogenetic analysis, these genes were grouped into two families. The alignment of the deduced amino acid sequences has allowed us to

arbitrarily divide the putative defensins of *M. truncatula* into three classes. Some of the defensins within Class I (BG452001, TC85327, CA990416 and TC89352) share excellent sequence homology near their carboxy terminus with the PDF2.1, PDF2.2, PDF2.3 and PDF2.4 defensins of *A. thaliana* (Thomma *et al.*, 2002). To our knowledge, Class II defensins have not been reported previously in a legume plant. The only plant defensin (Bp4A) that carries a deletion of two amino acids between the last two conserved cysteines has been reported to be a pollen-coating protein in *Brassica napus* (Albani *et al.*, 1990). It remains to be determined if the two defensins (TC76542 and TC87634) with a deletion of two amino acids between the last two conserved cysteines have a similar role in *M. truncatula*. The members within Class III share significantly greater sequence homology than those within Class I (Figure 3). With the exception of defensins represented by TC82368, TC87634 and TC90828, all defensins are predicted to carry a net positive charge ranging from +2 to +9. There are several defensins within Class I and Class III that carry a net charge of +6 or higher, some of which are

likely to have strong *in vitro* antifungal activity against plant fungi. We have already observed that MtDef1.1, which carries a net charge of +3, has strong *in vitro* antifungal activity against certain fungi, whereas MtDef2.1 (represented by TC82363), which carries no net positive charge, lacks antifungal activity against those same fungi (Spelbrink *et al.*, 2004).

It is now apparent that plant defensins are quite diverse in their amino acid composition and biological activity (Thomma *et al.*, 2002). A defensin protein isolated from flowers of *Nicotiana glauca* was recently shown to have not only antifungal activity, but also insecticidal activity against certain Lepidopteran insects. This protein inhibited the digestive enzymes trypsin and chymotrypsin of the insect gut (Lay *et al.*, 2003a, b). A defensin protein, Slα1, from Sorghum seed which lacks antifungal activity has been reported to inhibit insect gut α -amylase (Bloch and Richardson, 1991). At present, the possibility remains that defensins may have diverse biological functions in *M. truncatula*. A combination of biochemical, genetic and biological approaches will be required to decipher the biological functions of defensins in *M. truncatula*.

Our expression analysis indicates that, in *M. truncatula*, defensin genes are expressed in a variety of tissues during normal growth and development and are also induced in response to a variety of biotic and abiotic stimuli. The EST content of the individual TCs is derived from 2 to 18 libraries. Based on their large EST representation, four defensin genes (TC76542, TC77480, TC87273 and TC87634) are expressed at a moderate to high level although further validation of the expression level either by northern or macroarray hybridization is required. Some of these genes are expressed in leaves, flowers, roots and developing seeds. In addition, using northern blot analysis, we have determined that the MtDef1.1 (AJ498901) and MtDef2.1 (TC82368) genes are highly expressed in dry mature seed. High expression of these genes in seed strongly suggests a role for these proteins in defense against soil-borne pathogens during the early stages of seed germination. Strong support for a role of these proteins in defense during seed germination was suggested previously in radish where the seed defensins, Rs-AFP1 and Rs-AFP2, were shown to be released rapidly and preferentially during seed

imbibition (Terras *et al.*, 1995). None of the defensin ESTs come from any publicly searchable root cDNA library and our northern blotting data indicate little or no expression of the MtDef1.1 or MtDef2.1 gene in roots. Using RT-PCR, we have detected constitutive expression of the TC85327 gene in roots (Figure 5). Furthermore, expression of TC77480 gene is turned on moderately or highly in roots in response to mycorrhizal fungal infection or rhizobacterial infection. Whether or not these genes are playing a role during plant-bacterial or plant-fungal symbiosis remains to be determined. It is worth noting that a large number of nodule-specific transcripts have been recently shown to encode small cysteine-rich proteins in *M. truncatula* (Fedorova *et al.*, 2002; Mergaert *et al.*, 2003). It is speculated that these proteins could potentially serve as antimicrobial proteins preventing opportunistic infections by other soil microorganisms during nodule formation. It is conceivable that defensins could serve in similar capacity during nodule formation or mycorrhizal fungal infection of roots.

Though constitutively expressed in the seed, MtDef1.1 and MtDef2.1 genes are strongly induced in young seedlings by exogenous MeJA in *M. truncatula*, but not by ethylene or SA. In closely related *M. sativa*, however, up-regulation of these otherwise constitutively expressed genes is not observed by treatment with MeJA. Interestingly, constitutive expression of these genes is down-regulated by ethylene in *M. sativa*. The Arabidopsis defensin gene, PDF1.2, has been shown to be induced strongly by MeJA as well as ethylene, but not by SA (Penninckx *et al.*, 1996; Penninckx *et al.*, 1998). The data presented here suggest that at least some aspects of the regulation of defensin genes via MeJA and ethylene signaling pathways may be different not only between the crucifer (*A. thaliana*) and legumes, but also between the two closely related legumes.

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